

year, which is already expanding to 9000 metric tonnes. The recycling process starts with procuring e-waste, which is dismantled, segregated, and processed. For example, when recycling a fridge, the aluminium or plastic body is separated, and any oil or gas in the compressor is extracted. The PCBs are processed further to separate metals such as copper. For lithium-ion batteries, recycling involves shredding and separating materials like aluminium, copper, plastic, and black mass, which contains critical metals like lithium and cobalt.

What differentiates Recyclekaro's recycling process from traditional methods?

Traditional recycling methods use lower-temperature furnaces to melt precious metals like gold, silver, and platinum. However, rare earth metals like rhodium require incredibly high temperatures (5000-10,000 degrees Celsius), which can only be achieved using plasma furnace technology. We plan to implement plasma furnace technology to extract rare earth metals. While the current process uses hydrometallurgy (water- and chemical-based extraction) for most metals, the plasma furnace will enable the company to handle more complex materials that demand higher temperatures for effective recovery.

What is the role of plasma furnace technology in the recycling process?

Plasma furnace technology is crucial for extracting rare earth metals from e-waste, especially from components containing rare earth elements or rhodium. The plasma furnace heats materials to extreme temperatures (up to 10,000 degrees Celsius), melting the components into a composite material that is

later processed through hydrometallurgy. This high-temperature melting is essential for extracting rare earth metals that cannot be recovered through traditional lower-temperature methods.

What are the unique aspects of Recyclekaro's approach to battery recycling?

We use specialised machinery for lithium-ion battery recycling, including shredders and leaching reactors. Once the batteries are shredded, materials like aluminium, copper, and plastic are separated, while the black mass undergoes chemical processing to extract valuable metals like lithium, cobalt, and nickel. Our process is environmentally responsible and sustainable, maintaining a zero-discharge approach. Waste-water is treated in an evaporator plant, ensuring no harmful waste is released into the environment. This process allows us to produce industrial-grade metals like cobalt, which we supply to industries such as paint and fertilisers. Additionally, we are positioning ourselves to supply cathode-grade materials once battery manufacturing begins in India, preparing us for the industry's future.

How do you handle waste collection, and what is the scale of its operations?

We have established a pan-India network of collection centres in cities such as Delhi, Bengaluru, and Hyderabad. Through these centres, we collect e-waste and battery waste from a range of sources, including corporations and government auctions. For example, we have processed telephone exchanges from BSNL across India. Each collection centre is managed by a dedicated supervisor who oversees the process and ensures that waste is dismantled, segregated, and processed in

line with our high standards. This approach allows us to manage waste collection and recycling operations efficiently on a large scale across the country.

How do you ensure traceability and compliance with environmental regulations?

We strictly adhere to the Extended Producer Responsibility (EPR) guidelines set by the Central Pollution Control Board. Detailed traceability records are maintained for every load of waste processed. This involves photographing waste during loading and unloading, weighing trucks, and meticulously tracking the extraction process. All of this data is uploaded to the EPR portal, ensuring full compliance with regulatory standards.

How do you maintain quality control in your recycling processes?

Regular quality checks are conducted throughout the recycling process to ensure maximum efficiency. Solution values are tested regularly to maintain optimal process performance. Advanced machines are used to accurately assess the composition and purity of the extracted metals. Each batch of recovered metals undergoes rigorous testing, both in-house and by third-party laboratories, before dispatch. This guarantees that the metals meet the high standards required for industrial use, providing our clients with confidence in the quality and reliability of our materials.

What are the major challenges in the e-waste and battery recycling industry?

One of the biggest challenges in the recycling business is the lack of standardisation in the input material. For instance, e-waste like fridges contains only a small